WORKED SOLUTION

STEP III 1992, Question 14

Mechanics • Projectile motion

STEP III 1992, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s3-q14

The key idea

Everything is governed by the single angle θ. The free part of the string is tangent to the disc at Q, and its length is the arc of string that has come off the disc, so P is easy to locate in terms of θ (the curve traced is an *involute* of the circle). Differentiating, the velocity of P turns out to be perpendicular to the string. So the string's tension does no work, and only friction acts along the direction of motion. Resolving along and perpendicular to PQ then splits the problem neatly in two.

Solution

Position of P

Model P as a particle on a horizontal table whose only resistance is the given friction, and the string as light and inextensible. Gravity and the normal reaction cancel, so the only horizontal forces are the tension T along PQ and the friction.

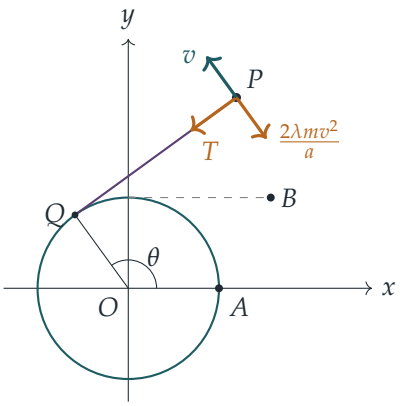
With $A = (a, 0)$ on the x-axis, the point of contact is $Q = (a\cos\theta,\, a\sin\theta)$. When P was at A, all the string was on the disc; since the string is inextensible, the free part QP has length equal to the arc AQ that has unwound, namely $a\theta$. The string is tangent at Q, so QP is perpendicular to OQ, in the direction of the unit vector $(\sin\theta,\, -\cos\theta)$ (at $\theta = \frac{\pi}{2}$ this is $(1, 0)$, pointing from the top of the disc towards B, as in the figure). Hence

$$\overrightarrow{OP} = a(\cos\theta,\, \sin\theta) + a\theta(\sin\theta,\, -\cos\theta),$$

so

$$x = a\cos\theta + a\theta\sin\theta, \qquad y = a\sin\theta - a\theta\cos\theta.$$

The first is the required x-coordinate. At B the free string is parallel to OA, touching the disc at the top, so $\theta = \frac{\pi}{2}$ there and $B = (\frac{\pi a}{2},\, a)$.



Velocity and acceleration

Write $\mathbf{e} = (\cos\theta, \sin\theta)$ for the unit vector along OQ and $\mathbf{f} = (-\sin\theta, \cos\theta)$ for the unit vector from P towards Q (the reverse of the string direction above). Note that $\dot{\mathbf{e}} = \dot\theta\,\mathbf{f}$ and $\dot{\mathbf{f}} = -\dot\theta\,\mathbf{e}$.

Differentiating x and y,

$$\dot x = (-a\sin\theta + a\sin\theta + a\theta\cos\theta)\,\dot\theta = a\theta\dot\theta\cos\theta,$$
$$\dot y = (a\cos\theta - a\cos\theta + a\theta\sin\theta)\,\dot\theta = a\theta\dot\theta\sin\theta,$$

so the velocity is $a\theta\dot\theta\,\mathbf{e}$: parallel to OQ, and so perpendicular to the string. Differentiating again with the product rule,

$$\ddot{\mathbf{r}} = a(\dot\theta^2 + \theta\ddot\theta)\,\mathbf{e} + a\theta\dot\theta\,\dot{\mathbf{e}} = a(\dot\theta^2 + \theta\ddot\theta)\,\mathbf{e} + a\theta\dot\theta^2\,\mathbf{f}.$$

So the acceleration has component $a\theta\dot\theta^2$ along PQ (towards Q) and $a\dot\theta^2 + a\theta\ddot\theta$ perpendicular to PQ (in the direction of $\mathbf{e}$), as required.

The equation of motion

At B, $\theta = \frac{\pi}{2}$ and the velocity is $a\cdot\frac{\pi}{2}\,\dot\theta\,(0,1)$. Projection perpendicular to OA and away from it means velocity V in the positive y-direction, so

$$\dot\theta = \frac{2V}{\pi a} > 0 \quad \text{initially.}$$

While $\dot\theta > 0$, the speed is $v = a\theta\dot\theta$ and the motion is in the direction $+\mathbf{e}$, so friction of magnitude $2\lambda mv^2/a = 2\lambda ma\theta^2\dot\theta^2$ acts in the direction $-\mathbf{e}$. The tension acts along $\mathbf{f}$ and has no $\mathbf{e}$-component.

> **Ask yourself** Which of the two acceleration components should we use to get an equation for θ alone?

The perpendicular one: the tension T is unknown, and it appears only in the component along PQ. Newton's second law perpendicular to PQ gives

$$m(a\dot\theta^2 + a\theta\ddot\theta) = -2\lambda ma\theta^2\dot\theta^2.$$

Dividing by $ma\theta\dot\theta$ (allowed, as $\theta > 0$ and $\dot\theta > 0$),

$$\frac{\ddot\theta}{\dot\theta} = -\left(\frac{1}{\theta} + 2\lambda\theta\right)\dot\theta,$$

which is the required result.

Finding $\dot\theta$

The left-hand side is $\frac{\mathrm{d}}{\mathrm{d}t}\ln\dot\theta$, and by the chain rule the right-hand side is $\frac{\mathrm{d}}{\mathrm{d}t}(-\ln\theta - \lambda\theta^2)$. Hence

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big(\ln(\theta\dot\theta) + \lambda\theta^2\Big) = 0, \qquad \text{so} \qquad \theta\dot\theta\,\mathrm{e}^{\lambda\theta^2} = \text{constant}.$$

At B the constant is $\frac{\pi}{2}\cdot\frac{2V}{\pi a}\,\mathrm{e}^{\lambda\pi^2/4} = \frac{V}{a}\,\mathrm{e}^{\lambda\pi^2/4}$. Therefore

$$\dot\theta = \frac{V}{a\theta}\,\exp\Big(\lambda\Big(\frac{\pi^2}{4} - \theta^2\Big)\Big).$$

This is positive for every θ, so $\dot\theta$ never vanishes and the assumption $\dot\theta > 0$ used above holds throughout. (The answer necessarily involves V as well as θ, λ and a.) Equivalently, the speed is $v = a\theta\dot\theta = V\mathrm{e}^{\lambda(\pi^2/4-\theta^2)}$.

Tension at $\theta = \pi$

Along PQ the only force is T, towards Q, so $T = ma\theta\dot\theta^2$. At $\theta = \pi$,

$$\dot\theta^2 = \frac{V^2}{a^2\pi^2}\,\mathrm{e}^{2\lambda(\pi^2/4-\pi^2)} = \frac{V^2}{a^2\pi^2}\,\mathrm{e}^{-3\lambda\pi^2/2},$$

so

$$T = ma\pi\cdot\frac{V^2}{a^2\pi^2}\,\mathrm{e}^{-3\lambda\pi^2/2} = \frac{mV^2}{\pi a}\,\mathrm{e}^{-3\lambda\pi^2/2}.$$

Checks

- The free string PQ has length $a\theta$ and is the radius of curvature of the involute, so Q is the centre of curvature and the component towards Q should be $v^2/(a\theta) = a\theta\dot\theta^2$. It is.
- With $\lambda = 0$ (no friction) the speed stays V, as it must, since the tension is perpendicular to the motion and does no work. At $\theta = \frac{\pi}{2}$, $T = mV^2/(\frac{\pi}{2}a)$, which is mv^2/r for the string length $r = \frac{\pi a}{2}$ at B.
- Numerically, with $a = 1.3$, $m = 0.7$, $V = 2.1$, $\lambda = 0.25$, we stepped the equation for θ with a Runge–Kutta method: it matched the formula for $\dot\theta$ to 10^{-14}. Finite-difference accelerations of (x, y) matched both components and the friction, and gave $T \approx 0.018667$ at $\theta = \pi$, agreeing with the formula.

Common slips

- Using $\theta = 0$ as the starting value: the particle starts at B, where $\theta = \frac{\pi}{2}$, not at A.
- Resolving along PQ to find $\dot{\theta}$: that equation contains the unknown tension, and friction has no component there.
- Forgetting that friction opposes the velocity, which is along OQ, not along the string.

Beyond STEP

The path of P is the involute of a circle, the curve traced by the end of a taut string unwinding from a spool, and the shape used for the teeth of most gears. The facts used here (the velocity is perpendicular to the string, and the string's length is the radius of curvature) are special cases of the evolute–involute relationship in the differential geometry of curves. Dynamically, it is a good example of using intrinsic (tangential and normal) components to separate a known force from an unknown constraint force. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats motion in polar and intrinsic coordinates and constrained motion in this spirit.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.